# Identifying Key Indicators of Spam Emails: A GLM Approach

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# Background

Which text characteristics influence whether an email will be classified as spam or not?

Objective: Optimize Spam Filters

# Data Cleaning

```
# A tibble: 6 × 7
  crl.tot dollar  bang money  n000  make yesno
    <dbl>  <dbl> <dbl> <dbl> <dbl> <dbl> <chr>
1     253  0.189 0.882  0     0.37  0    y
2     112  0.08  0      0     0     0    n
3      17  0     0.5    0     0     0    y
4     319  0     0.141  0     0     0    n
5     513  0.464 0.406  0     0.66  0.33 y
6    2049  0.157 0.502  0.08  0.42  0    y
```

Response (Y): yes/no (Spam vs. Non-spam)
Predictors (X): 6 text-based features

Data summary

| | |
|---|---|
| Name | spam_clean |
| Number of rows | 920 |
| Number of columns | 7 |
| _____ | |
| Column type frequency: | |
| factor | 1 |
| numeric | 6 |
| _____ | |
| Group variables | None |

**Variable type: factor**

| skim_variable | n_missing | complete_rate | ordered | n_unique | top_counts |
|---|---|---|---|---|---|
| yesno | 0 | 1 | FALSE | 2 | n: 557, y: 363 |

**Variable type: numeric**

| skim_variable | n_missing | complete_rate | mean | sd | p0 | p25 | p50 | p75 | p100 | hist |
|---|---|---|---|---|---|---|---|---|---|---|
| crl.tot | 0 | 1 | 265.76 | 502.61 | 1 | 35.75 | 89.5 | 240.50 | 3498.00 | ▇▁▁▁ |
| dollar | 0 | 1 | 0.07 | 0.26 | 0 | 0.00 | 0.0 | 0.03 | 5.30 | ▇▁▁▁ |
| bang | 0 | 1 | 0.24 | 0.55 | 0 | 0.00 | 0.0 | 0.30 | 7.84 | ▇▁▁▁ |
| money | 0 | 1 | 0.07 | 0.30 | 0 | 0.00 | 0.0 | 0.00 | 6.66 | ▇▁▁▁ |
| n000 | 0 | 1 | 0.09 | 0.33 | 0 | 0.00 | 0.0 | 0.00 | 4.32 | ▇▁▁▁ |
| make | 0 | 1 | 0.12 | 0.36 | 0 | 0.00 | 0.0 | 0.00 | 4.34 | ▇▁▁▁ |

# Exploratory Data Analysis (EDA)

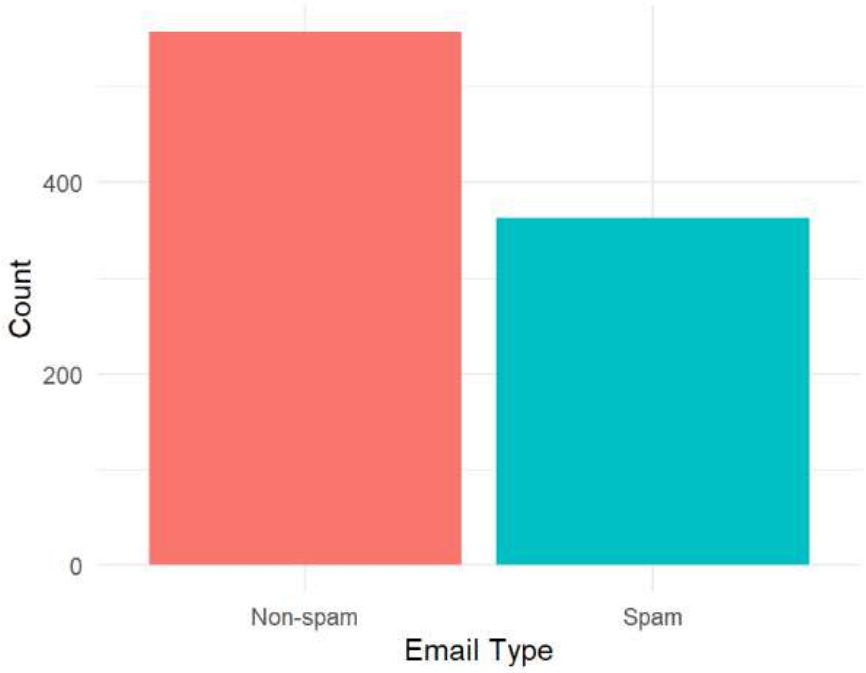



Table 1: Grouped summary of predictors by email type.

| yesno | variable | mean | median | zero_pct | max |
|---|---|---|---|---|---|
| n | bang | 0.074 | 0.000 | 75.9% | 3.571 |
| n | crl.tot | 168.259 | 55.000 | 0.0% | 3467.000 |
| n | dollar | 0.010 | 0.000 | 89.6% | 0.427 |
| n | make | 0.101 | 0.000 | 84.2% | 4.340 |
| n | money | 0.005 | 0.000 | 98.7% | 0.800 |
| n | n000 | 0.006 | 0.000 | 97.1% | 0.960 |
| y | bang | 0.490 | 0.298 | 20.1% | 7.843 |
| y | crl.tot | 415.372 | 171.000 | 0.0% | 3498.000 |
| y | dollar | 0.158 | 0.051 | 43.8% | 5.300 |
| y | make | 0.145 | 0.000 | 67.8% | 2.350 |
| y | money | 0.177 | 0.000 | 66.4% | 6.660 |
| y | n000 | 0.219 | 0.000 | 71.1% | 4.320 |

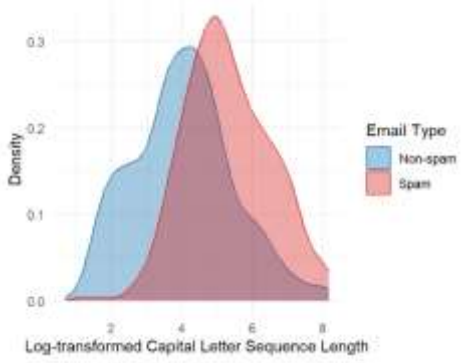
Sample Size:

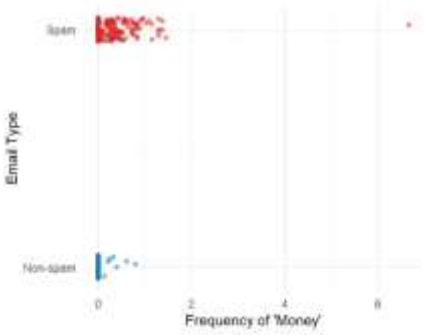
Total Observations: n = 920

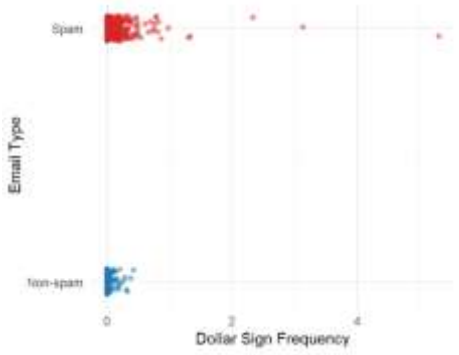
Non-Spam: $n_1$ = 557 (60.5%)

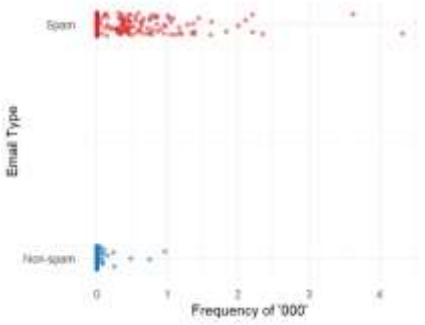
Spam: $n_2$ = 363(39.5%)

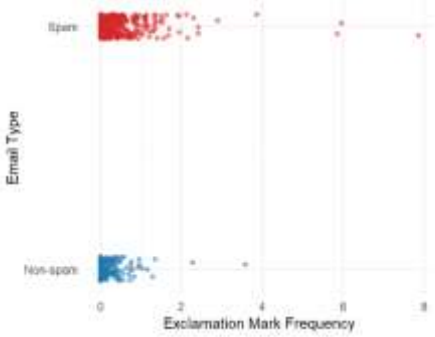
# Exploratory Data Analysis (EDA) - Key Findings

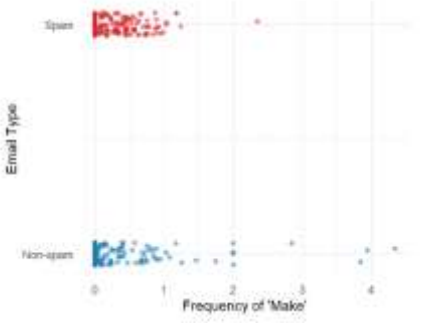


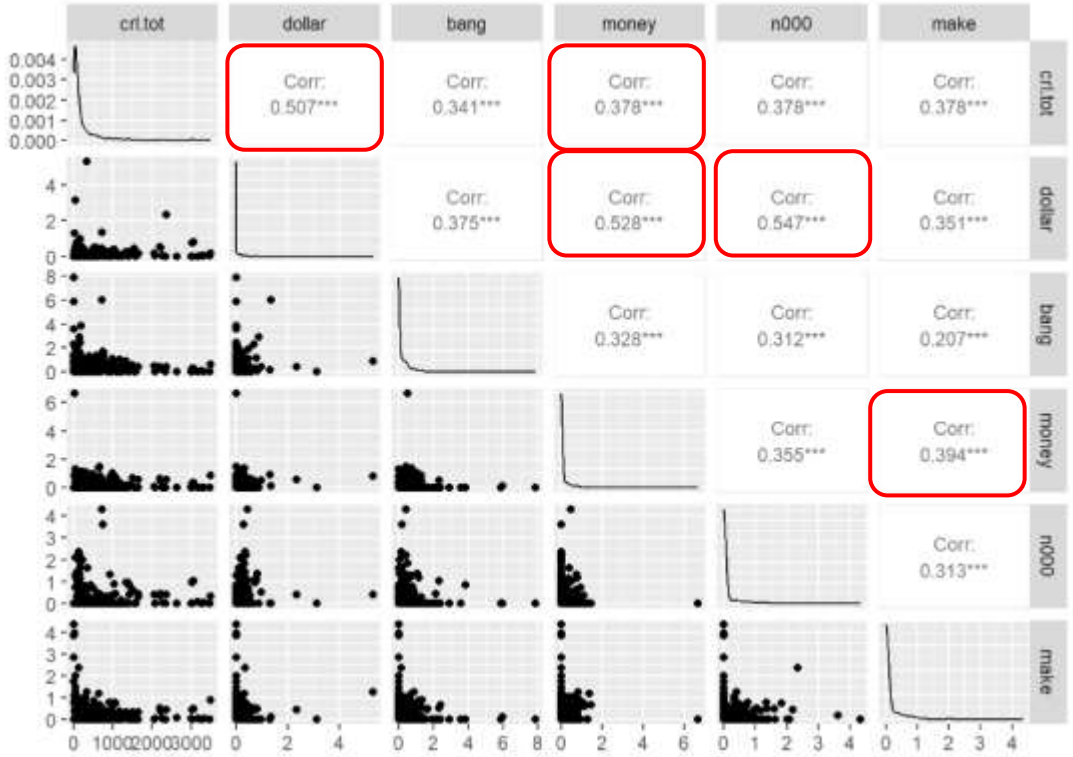




Spam Indicators: Strong patterns observed in **dollar, bang,** and **money**.

Key Feature: String **n000** shows a distinct frequency spike in spam.

Weak Pattern: Variable **make** appears less discriminative.

Model Suitability: Low multicollinearity; variables ready for Logistic Regression.

# Statistical Analysis: Logistic Regression

$$\ln\left(\frac{\mathrm{p}}{1-\mathrm{p}}\right) = \beta_0 + \beta_1\ (\mathrm{crl.tot}) + \beta_2(\mathrm{dollar}) + \beta_3(\mathrm{bang}) + \beta_4(\mathrm{money}) + \beta_5(\mathrm{n000}) + \beta_6(\mathrm{make})$$

- p is the probability that the email is spam.
- $\ln\left(\frac{\mathrm{p}}{1-\mathrm{p}}\right)$ is known as the log-odds.
- $\beta_0$ is the intercept, and $\beta_1 \ldots \beta_6$ are the regression coefficients for each variable.

# Model Fitting

```
Call:
glm(formula = yesno ~ crl.tot + dollar + bang + money + n000 +
    make, family = binomial(link = "logit"), data = spam_clean)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.8101190  0.1254636 -14.427  < 2e-16 ***
crl.tot      0.0005502  0.0001886   2.917 0.003533 **
dollar       8.1346140  1.5396484   5.283 1.27e-07 ***
bang         2.9172085  0.3363971   8.672  < 2e-16 ***
money        5.9724851  1.2455257   4.795 1.63e-06 ***
n000         3.4827736  1.0261134   3.394 0.000688 ***
make        -0.4553154  0.4065463  -1.120 0.262731
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1234.17  on 919  degrees of freedom
Residual deviance:  752.44  on 913  degrees of freedom
AIC: 766.44

Number of Fisher Scoring iterations: 7
```

# Model Selection & Results

**Full : yesno ~ crl.tot + dollar + bang + money + n000 + make**

**Reduced : yesno ~ crl.tot + dollar + bang + money + n000**

Selection: Reduced model via backward elimination.

- Full Model AIC: 766.44
- Reduced Model AIC: 766.04

Improved parsimony without sacrificing predictive power.

| | OR | 2.5 % | 97.5 % |
|---|---|---|---|
| (Intercept) | 0.158 | 0.123 | 0.200 |
| crl.tot | 1.001 | 1.000 | 1.001 |
| dollar | 3573.436 | 202.301 | 85411.590 |
| bang | 18.819 | 10.013 | 37.592 |
| money | 341.550 | 38.501 | 5244.916 |
| n000 | 30.794 | 5.218 | 301.686 |

| | Actual | |
|---|---|---|
| Predicted | n | y |
| n | 523 | 122 |
| y | 34 | 241 |

- Overall Accuracy: 83.04 %
- False Positive Rate: 6.1 %
- Sensitivity: 67.4%

**dollar**, **money** are highly significant.
**n000**, **bang** are significant.
**Capital letters** have a negligible effect on predictions, or possibly no effect at all.

# Model Validation & Performance

## 80/20 Train-Test

```
Call:
glm(formula = yesno ~ crl.tot + dollar + bang + money + n000,
    family = binomial(link = "logit"), data = spam_train)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.0268297  0.1439189 -14.083  < 2e-16 ***
crl.tot      0.0005845  0.0001947   3.002  0.00268 **
dollar       8.6024844  1.8841848   4.566 4.98e-06 ***
bang         3.1467016  0.3877579   8.115 4.85e-16 ***
money        8.6141664  1.9143086   4.500 6.80e-06 ***
n000         2.7695467  1.0437369   2.653  0.00797 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 968.57  on 735  degrees of freedom
Residual deviance: 565.88  on 730  degrees of freedom
AIC: 577.88

Number of Fisher Scoring iterations: 8
```

| | Actual | |
|---|---|---|
| Predicted | n | y |
| n | 85 | 35 |
| y | 7 | 57 |

Final Test Accuracy: 77.17 %
False Positive Rate (Unseen Data): 7.61 %

# Conclusion, Future Work & Limitations

Spam and non-spam emails showed clear differences in financial, promotional, and emphatic text features.

Strongest predictors:

**dollar** ($P < 0.001$, OR ≈ 3573), **money** ($P < 0.001$, OR ≈ 341)

Moderate predictors:

**n000** ($P < 0.01$, OR ≈ 30.8), **bang** ($P < 0.001$, OR ≈ 18.8)

Weak practical effect:

**crl.tot** was statistically significant, but OR ≈ 1.001, so its practical effect was very small.

make was removed from the reduced model, which gave a simpler model while keeping the main useful predictors.

**83% Accuracy** confirms text-based features are effective for real-world baseline filtering.

# Conclusion, Future Work & Limitations

## Limitations:

The model used only a limited number of text-based predictors. And some predictors (e.g., crl.tot) were statistically significant but had weak practical effects, so The model should be seen as a useful baseline rather than a final solution.

## Future Work:

Use a larger and more diverse dataset.
Add more text-based features.
Try other classification methods.

# Q & A

Thanks for your listening!